Pharmacognosy I
CPG 202

Seeds
(Evening primrose + Horse chestnut + Pumpkin + Colchicum)

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Learning outcomes

a1	Describe the main morphological and histological characters of drugs composed of flowers, seeds, fruits, in addition to drugs of marine and animal origin.
a2	Outline the active constituents, folk and clinical proven uses of selected drugs composed of flowers, seeds, fruits or marine and animal origin, their pharmacological actions, adverse effects, drug interactions and available phytopharmaceuticals.
a3	Identify official drugs belonging to flowers, seeds, fruits, in addition to drugs of marine and animal origin and their adulterants using morphological and microscopical characters or chemical tests per their reported monographs.
c1	Relate medicinal uses, actions and adverse effects of herbal drugs based on their content of secondary metabolites.
c2	Select a suitable botanical drug for treatment of a certain disease.

Evening primrose بذور حشيشة الحمار

The dried seeds of *Oenothera biennis*
Family: Onagraceae

Synonyms: Enotera & la belle de nuit
زهرة الربيع المسائية

Mainly used as fixed oil obtained from the seeds

Description

- Dark colored
- Irregular
- Very small in size

Major Constituents

- **Fixed oil (14%) EPO**
  - ➢ ***γ*- Linolenic acid**
  - ➢ ***Cis*- Linoleic acid**
  - ➢ **Oleic acid**

Essential fatty acids

EVENING PRIMROSE OIL

Supplement Facts

Serving Size: 1 Softgel

Amount Per Softgel		**% Daily Value**
Calories	10	
Calories from Fat	10	
Total Fat	1 g	2%
Saturated Fat	0 g	0%
Polyunsaturated Fat	1 g	*
Organic Evening Primrose Seed Oil		*
(Cold-Pressed)	1.3 g	
Supplying approximately:		
Gamma-Linolenic acid (GLA)	76 mg	
Linoleic acid	544 mg	
Oleic acid	49.9 mg	

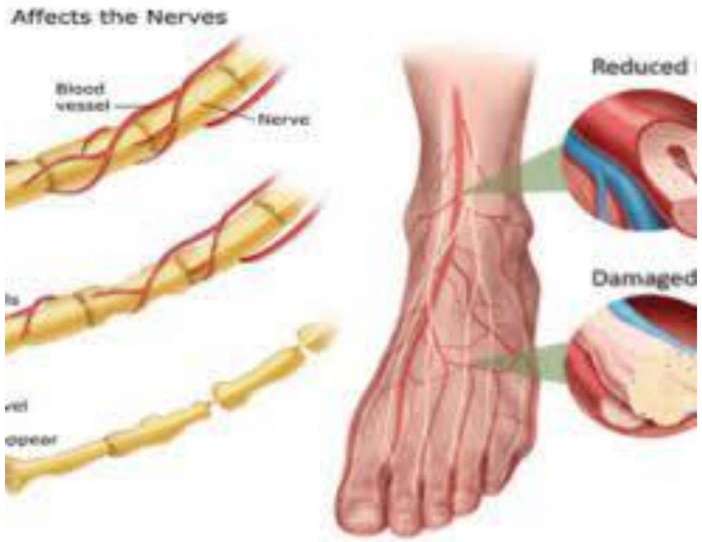
Percent Daily Values are based on a 2,000 calorie diet.
*Daily Value not established.

Other ingredients: Gelatin, glycerin, and water.

Free of the following common allergens: milk/casein, eggs, fish, shellfish, tree nuts, peanuts, wheat/gluten, corn, yeast, and soybeans. Contains no artificial colors, flavors or preservatives.

Medicinal Uses

- Atopic eczema
- Diabetic neuropathy
- Menstrual cramps
- Premenstrual syndrome (PMS)



Mechanism of action

γ- Linolenic acid — Inactive form

↓

Dihomo-γ- Linolenic acid — Active form

↓

PGE 1
(anti-inflammatory mediators) ↑

Safety

- Safe for internal and external use

Drug interactions

- Decrease platelets aggregation, contraindicated with the use of anticoagulants

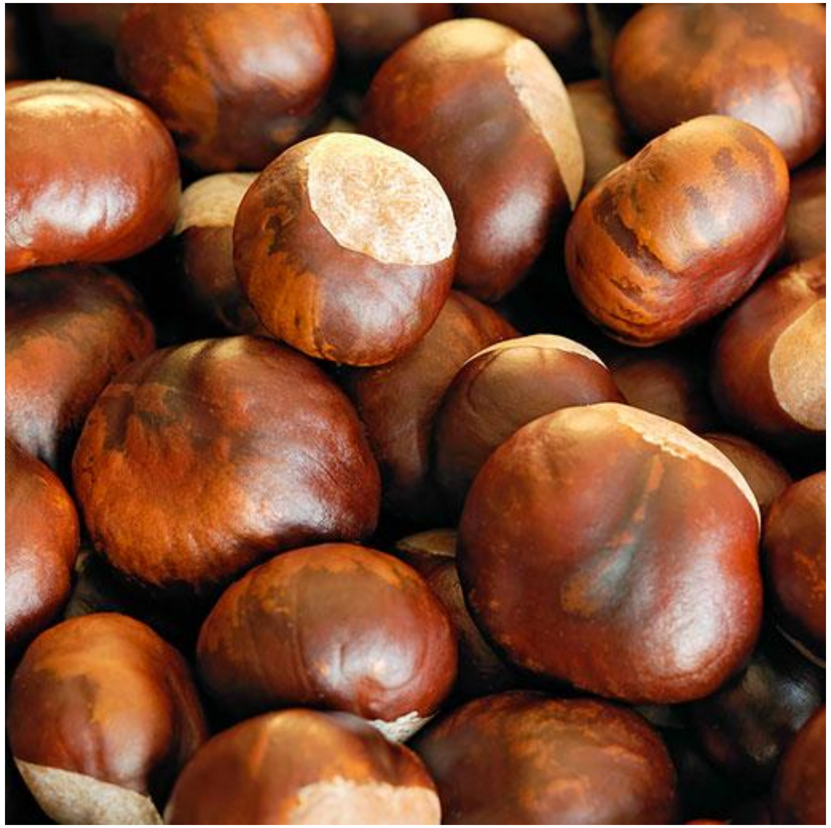
Horse Chestnut أبو فروة الحصان

The dried ripe seeds of *Aesculus hippocastanum* Family: Hippocastananceae

Synonyms: Castagna amara & أبو فروة

Description

- Ovoid or globular
- Shiny hard testa
- Large hilum
- Large fleshy oily cotyledons



Major Constituents

- Triterpene saponins
  - Aescin
- Flavonoids
- Tannins
- Fixed oils & fats

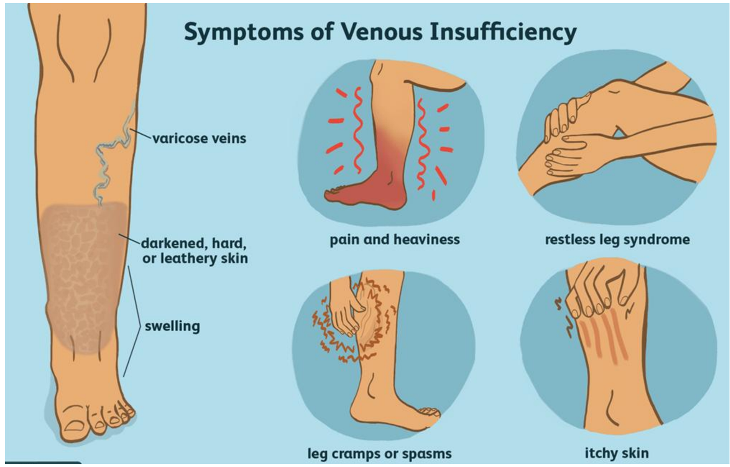
Medicinal Uses

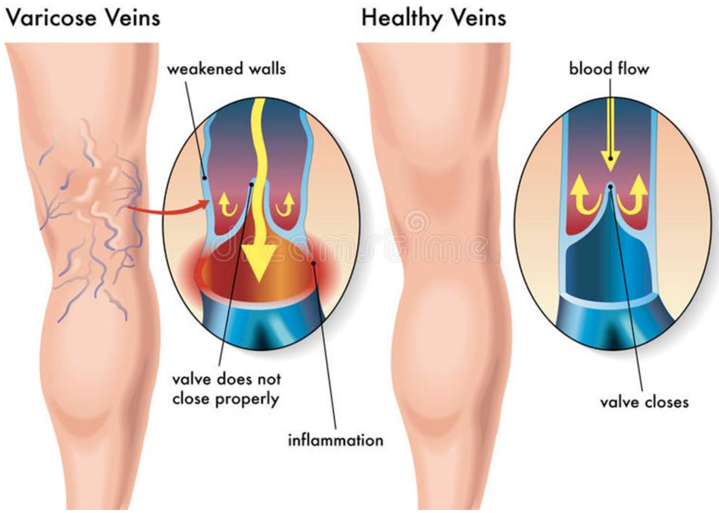
- **Internally**

➢ **Venous insufficiency**

➢ **Oedema**

➢ **Heaviness in legs**



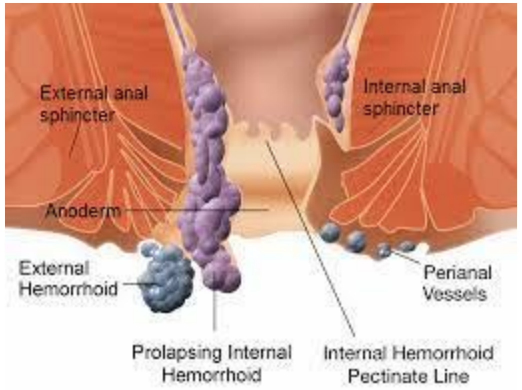


- **Externally**

➢ **Varicose veins**

➢ **Hemorrhoids**

➢ **Sprains, bruises, and haematoma**



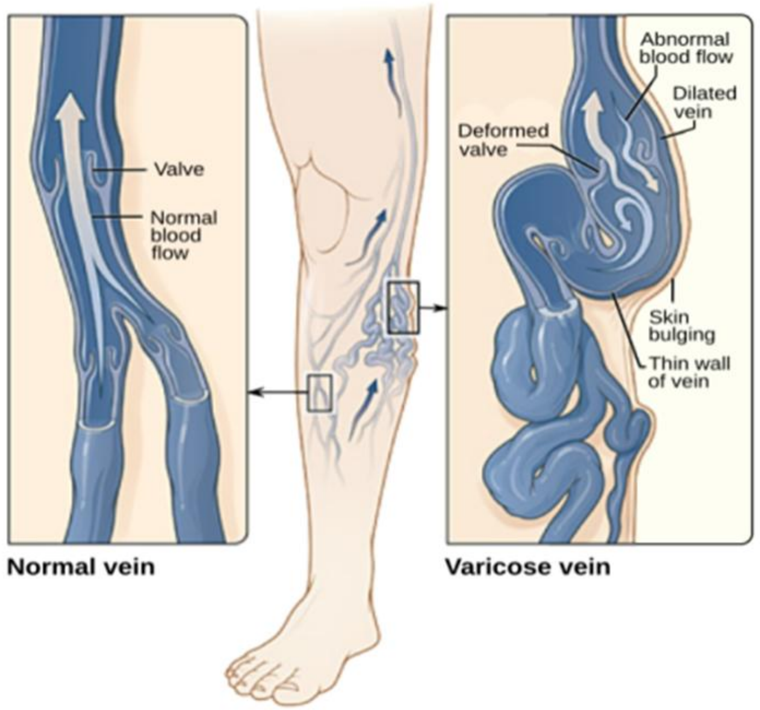
Mechanism of action

Aescin & flavonoids

→ Venotonic

→ Anti-inflammatory

→ Decrease vascular permeability



Pumpkin seeds بذور القرع

The dried seeds of *Cucurbita pepo* L. and its cultivars Family: Cucurbitaceae

Synonyms: Pompion بذر القرع



Native to America and cultivated worldwide

Description

- ➢ **Ovate & flat**
- ➢ **Creamy white testa**
- ➢ **Exalbuminous**
- ➢ **Anatropous ovule**
- ➢ **Straight dicot embryo**

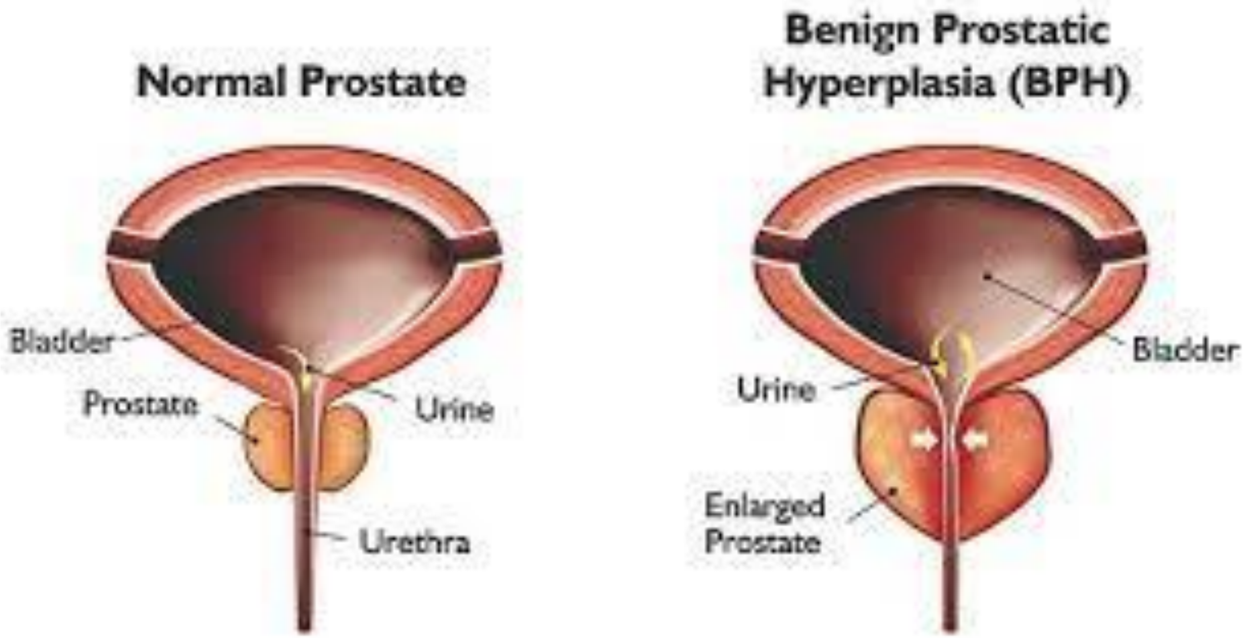
Major Constituents

- **Fixed oil**
  - Linoleic acid
  - α-Linolenic acid (ω-3)
  - Oleic acid
  - Squalene
- **Phytosterols**
- **Amino acids (Rare amino acids)**
  - Cucurbitine
- **Carotenoids**

Medicinal Uses

- **Symptoms & complications of benign prostatic hyperplasia (BPH)**

  - **Frequent urination**
  - **Irritation during urination**



Mechanism of action

Phytosterol

- Androgens antagonist
- Inhibits 5^{α}-reductase (Inhibits the synthesis of dihydrotestosterone)

α-Linolenic acid (ω-3)

- Anti-inflammatory (synthesis of PGE 3 anti-inflammatory mediators)

Colchicum seeds بذور اللحلاح

The dried seeds of *Colchicum autumnale*
Family: Liliaceae

Synonyms: Autumn crocus & Meadow saffron seed

Grows wild through Europe

Major Constituents

- Alkaloid
  - Colchicine
- Resin
- Fixed oil

Medicinal Uses

- **Anti inflammatory & analgesic especially in gout**

Mechanism of action

- **Colchicine blocks the inflammatory response to deposition of uric acid crystals**

Safety

- **Colchicine doses should be controlled because it is cytotoxic in large doses.**
- **People suffering from kidney diseases, liver diseases or cardiovascular diseases should avoid using colchicine.**

Adverse reaction

- Diarrhea, nausea, & vomiting
- Leukopenia & aplastic anemia
- Alopecia